

Exp.No.8-Malware Detection and Removal Simulation in Kali Linux

Aim

To simulate a malware infection, detect the malicious file, and safely remove it as part of the eradication phase of incident handling.

Objective

- To understand the malware eradication process
- To simulate a suspicious/malicious file
- To detect malicious files using system commands
- To remove malicious files safely

```
(unknown@kali):~$ touch malware.sh
(unknown@kali):~$ echo "echo malicious code running " >> malware.sh
(unknown@kali):~$ chmod +x malware.sh
(unknown@kali):~$ ./malware.sh
malicious code running
(unknown@kali):~$ ls -l
total 80
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Desktop
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Documents
drwxr-xr-x 2 unknown unknown 4096 Aug 19 09:52 Downloads
drwxrwxr-x 2 unknown unknown 4096 Aug 20 21:24 False_Positive_Lab
drwxrwxr-x 2 unknown unknown 4096 Aug 22 11:20 FIM_LAB
-rw-rw-r-- 1 unknown unknown 210 Jul 26 10:29 hid
drwxrwxr-x 3 unknown unknown 4096 Jul 26 09:21 Intran
drwxrwxr-x 2 unknown unknown 4096 Aug 20 21:54 IOCS
-rw-r--r-- 1 unknown unknown 4 Jul 28 17:57 lesson3.txt
-rwxrwxr-x 1 unknown unknown 29 Aug 22 13:43 malware.sh
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Music
-rw-rw-r-- 1 unknown unknown 247 Jul 27 10:42 one.txt
drwxrwxr-x 2 unknown unknown 4096 Aug 20 21:42 PMSHING
drwxr-xr-x 2 unknown unknown 4096 Aug 22 13:25 Pictures
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Projects
drwxr-xr-x 2 unknown unknown 4096 Jul 26 09:23 Public
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Templates
drwxrwxr-x 2 unknown unknown 4096 Jul 22 11:14 test1
drwxrwxr-x 2 unknown unknown 4096 Aug 3 13:22 test2
-rw-rw-r-- 1 unknown unknown 0 Aug 20 14:30 test.txt
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Videos

(unknown@kali):~$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.0 26804 16404 ?        Ss   13:16   0:01 /sbin/init splash
root           2  0.0  0.0  0 0 ?        S    13:16   0:00 [kthreadd]
root           3  0.0  0.0  0 0 ?        S    13:16   0:00 [pool_workqueue_release]
root           4  0.0  0.0  0 0 ?        I<   13:16   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0  0 0 ?        I<   13:16   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0  0 0 ?        I<   13:16   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0  0 0 ?        I<   13:16   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0  0 0 ?        I<   13:16   0:00 [kworker/R-netns]
root          10  0.0  0.0  0 0 ?        I<   13:16   0:00 [kworker/0:0M-kblockd]
root          12  0.0  0.0  0 0 ?        I    13:16   0:00 [kworker/u8:0-ipv6_addrconf]
root          13  0.0  0.0  0 0 ?        I<   13:16   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0  0 0 ?        S    13:16   0:00 [ksoftirqd/0]
root          15  0.0  0.0  0 0 ?        I    13:16   0:01 [rcu_preempt]
root          16  0.0  0.0  0 0 ?        S    13:16   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          17  0.0  0.0  0 0 ?        S    13:16   0:00 [rcu_exp_gp_kthread_worker]
```

```

unknown 2013 0.0 0.5 483880 20212 ? Ssl 13:17 0:00 /usr/libexec/xdg-desktop-portal
unknown 2020 0.0 0.1 308876 6996 ? Ssl 13:17 0:00 /usr/libexec/xdg-permission-store
unknown 2025 0.0 0.1 539960 7536 ? Ssl 13:17 0:00 /usr/libexec/xdg-document-portal
root 2034 0.0 0.0 2604 2100 ? Ss 13:17 0:00 fusermount3 -o rw,nosuid,nodev,fsname=portal,auto_unmount,subtype=portal
unknown 2040 0.0 0.5 483996 20920 ? Ssl 13:17 0:00 /usr/libexec/xdg-desktop-portal-gtk
unknown 4576 0.5 1.8 667656 75264 ? Sl 13:21 0:07 /usr/bin/qterminal
unknown 4584 0.2 0.1 10552 6832 pts/0 Ss 13:21 0:02 /usr/bin/zsh
root 8859 0.0 0.0 0 0 ? I< 13:29 0:00 [kworker/1:2H]
root 11622 0.0 0.0 0 0 ? I 13:35 0:00 [kworker/0:0-events]
root 12228 0.0 0.0 0 0 ? I 13:36 0:00 [kworker/1:0-events]
root 13253 0.1 0.0 0 0 ? I 13:38 0:00 [kworker/1:3-ata_sff]
root 13499 0.0 0.0 0 0 ? I 13:39 0:00 [kworker/u10:2-events_unbound]
root 15337 0.0 0.1 9740 4948 ? S 13:42 0:00 systemd-userwork: waiting...
root 15462 0.0 0.1 9740 5016 ? S 13:42 0:00 systemd-userwork: waiting...
root 15520 0.0 0.1 9740 4952 ? S 13:42 0:00 systemd-userwork: waiting...
root 15901 0.0 0.0 0 0 ? I 13:43 0:00 [kworker/1:1-ata_sff]
unknown 16131 0.0 0.1 9640 4528 pts/0 R+ 13:44 0:00 ps aux

```

```

(unknown@kali)-[~]
$ rm malware.sh

```

```

(unknown@kali)-[~]
$ ls -l

```

```

total 76
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Desktop
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Documents
drwxr-xr-x 2 unknown unknown 4096 Aug 19 09:52 Downloads
drwxrwxr-x 2 unknown unknown 4096 Aug 20 21:24 False_Positive_Lab
drwxrwxr-x 2 unknown unknown 4096 Aug 22 11:20 FIM_LAB
-rw-rw-r-- 1 unknown unknown 210 Jul 26 10:29 hid
drwxrwxr-x 3 unknown unknown 4096 Jul 26 09:21 imran
drwxrwxr-x 2 unknown unknown 4096 Aug 20 21:54 IOC5
-r----- 1 unknown unknown 4 Jul 28 17:57 lesson3.txt
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Music
-rw-rw-r-- 1 unknown unknown 247 Jul 27 10:42 one.txt
drwxrwxr-x 2 unknown unknown 4096 Aug 20 21:42 PHISHING
drwxr-xr-x 2 unknown unknown 4096 Aug 22 13:35 Pictures
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Projects
drwxr-xr-x 2 unknown unknown 4096 Jul 26 09:23 Public
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Templates
drwxrwxr-x 2 unknown unknown 4096 Jul 22 11:14 test1
drwxrwxr-x 2 unknown unknown 4096 Aug 3 13:22 test2
-rw-rw-r-- 1 unknown unknown 0 Aug 20 14:30 test.txt
drwxr-xr-x 2 unknown unknown 4096 Jul 17 17:48 Videos

```